

3.3 Battery Types

The following should give you an idea of the different types of batteries for mobile phones.

Nickel Cadmium (NiCd): These batteries are the least expensive of rechargeable batteries, but they suffer from inherent shortcomings - for instance, the memory effect. Here, if the battery is not fully drained before you recharge it, the unused portion eventually crystallises and hence, the complete capacity of the battery is not fully utilised. These types of batteries are no longer used in new mobile phone models.



A standard battery for a mobile phone

Nickel Metal Hydride (NiMH): This technology offers between 30 and 40 per cent longer operation as compared to NiCd batteries, but is susceptible to loss in performance at higher temperatures. These, too, are prone to the memory effect, but to a significantly lesser extent as compared to NiCd batteries. NiMH batteries have also been phased out on mobile phones.

Lithium Ion (Li-ion): The prime advantage that these batteries have over the older NiMH batteries is that they hold more energy for the same weight and volume. Therefore, you can get the same amount of power from much smaller and thinner batteries. Also, they do not have the inherent memory effect disadvantage of NiMH batteries. The only disadvantage with these batteries is that they cost more than other types. Most new mobile phones use Li-ion batteries.

Lithium Ion Polymer (Li-Po): Lithium ion polymer batteries, or more commonly, lithium polymer batteries, are rechargeable bat-